Abhinav Narayan Harish

CONTACT INFORMATION

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EDUCATION

University of Wisconsin–Madison (08/2023 – 12/2028, expected)

- Ph.D. in Computer Science.
- Research focus: Hallucinations in generative models.
- Advisor: Dr. Grigoris Chrysos.

Georgia Institute of Technology (08/2021 – 08/2023)

- M.S. in Electrical and Computer Engineering. GPA: 4.0/4.0.
- Thesis: Skill-Chaining for Long-Horizon Rearrangement [PDF].
- Advisor: Dr. Zsolt Kira.

Indian Institute of Technology Gandhinagar (07/2016 – 08/2020)

- Bachelor's in Electrical Engineering, minor in Computer Science. GPA: 9.41/10.
- **Institute Silver Medal**.
  - Ranked 2nd out of 34.

ACADEMIC RESEARCH EXPERIENCE

Georgia Institute of Technology, Atlanta, GA, USA
Graduate Research Assistant — **Jan. 2022 – May 2022**

- Advisor: Dr. Zsolt Kira.
- Set up a Habitat 2.0 simulation environment around a reconstructed Hubble Telescope scene to support research on remote repair and maintenance tasks.
- Assisted in testing LoCoBot navigation developed by the planning team, focusing on integrating sensing for obstacle detection and control for obstacle avoidance.

Indian Institute of Technology Gandhinagar, Gujarat, India
Research Fellow — **Aug. 2020 – Aug. 2021**

- Advisor: Dr. Shanmuganathan Raman.
- Formulated autonomous assembly of 3D objects as sequential shape assembly from part point clouds.
- Developed a bi-directional RNN integrated with a graph neural network to capture both top-down assembly structure and bottom-up part relationships.
- Improved part placement accuracy by 10% and inter-part connectivity prediction by 15% over prior baselines on PartNet.

INDUSTRY RESEARCH EXPERIENCE

NVIDIA Corporation, Santa Clara, CA, USA
Research Intern — **May 2022 – Aug. 2022**

- Mentor: Dr. Jonathan Tremblay.
- Built a Sim2Real workflow for object detection using Detectron2, generating synthetic training data and validating models in Omniverse Replicator.
- Studied how domain randomization over lighting, pose, camera view, and backgrounds improves model robustness and generalization.

Awards and Honors

- Member of WisTex United, winners of the RoboCup SPL Challenge Shield (2024).
- Summer RA Award, UW–Madison, for research with Dr. Josiah P. Hanna.
- Sabarmati Bridge Fellowship, IIT Gandhinagar; one of 9 recipients (2020–21).
- Best Poster Award (of ≈75 posters), Summer Research Internship Program, IIT Gandhinagar (2017).
- KVPY Fellowship (Kishore Vaigyanik Protsahan Yojana), Government of India; ranked 420th of ≈20,000 applicants.

Publications (* = equal contribution)

Peer-reviewed Conference Papers

4. M. H. Ashiq, S. Arora, **A. N. Harish**, I. Kharbanda, H. Y. Tseng, and G. Chrysos. Why DDIM Hallucinates More than DDPM: A Theoretical Analysis of Reverse Dynamics. International Conference on Machine Learning (ICML), 2026. [PDF].
3. A. Labiosa, Z. Wang, S. Agarwal, W. Cong, G. Hemkumar, **A. N. Harish**, B. Hong, J. Kelle, C. Li, Y. Li, Z. Shao, P. Stone, and J. P. Hanna. Reinforcement Learning Within the Classical Robotics Stack: A Case Study in Robot Soccer. International Conference on Robotics and Automation (ICRA), 2025. [PDF].
2. **A. N. Harish**, L. Heck, J. P. Hanna, Z. Kira, and A. Szot. Reinforcement Learning via Auxiliary Task Distillation. European Conference on Computer Vision (ECCV), 2024. [PDF].
1. **A. N. Harish**, R. Nagar, and S. Raman. RGL-NET: A Recurrent Graph Learning Framework for Progressive Part Assembly. Winter Conference on Applications of Computer Vision (WACV), 2022. [PDF].

Peer-reviewed Workshop Papers

2. J. P. Hanna, B. Pavse, and **A. N. Harish**. Replacing Implicit Regression with Classification in Policy Gradient Reinforcement Learning. Finding the Frame Workshop, Reinforcement Learning Conference (RLC), 2024. [PDF].
1. **A. N. Harish***, V. Verma*, and N. Khanna. Double JPEG Compression Detection of Distinguishable Blocks in Images Compressed with Same Quantization Matrix. International Workshop on Machine Learning for Signal Processing (MLSP), 2020. [PDF].

Service and Teaching

Reviewing

- International Conference on Learning Representations (ICLR) 2027.
- International Journal of Computer Vision (IJCV).
- ICLR 2026 Workshop: Agentic AI in the WILD: From Hallucinations to Reliable Autonomy.

Mentoring

- Muhammad Ashiq. Next: Ph.D. student at the University of Michigan.
- Samanyu Arora.
- Pranav Mehrotra.
- Eugene Kim.
- Jeffrey Yao.

Teaching Assistant

University of Wisconsin–Madison, Madison, WI, USA
Introduction to Artificial Intelligence (CS 540) **Fall 2024 – Spring 2026**
Data Science and Programming I (CS 220) **Aug. 2024 – May 2025**

Georgia Institute of Technology, Atlanta, GA, USA
Deep Learning (CS 7643) **Aug. 2022 – Aug. 2023**

Indian Institute of Technology Gandhinagar, Gujarat, India
Machine Learning (ES 654) **Jan. 2019 – May 2019**

Technical Skills

- Languages: Python, C++.
- Frameworks/Libraries: PyTorch, Gymnasium, Stable-Baselines3, NumPy, SciPy, OpenCV.
- Simulators: Habitat 2.0, PyGame.